CARTA DE APROXIMACIÓN
POR INSTRUMENTOS-OACI

ELEV AD
52 ft
VAR 1°W (2020)

APP 125.955 C
TWR E 118.780 C
GMC E 121.955 C
ATIS ARR 120.380 C

**MÁLAGA/Costa del Sol**
ILS Y
**RWY 12**

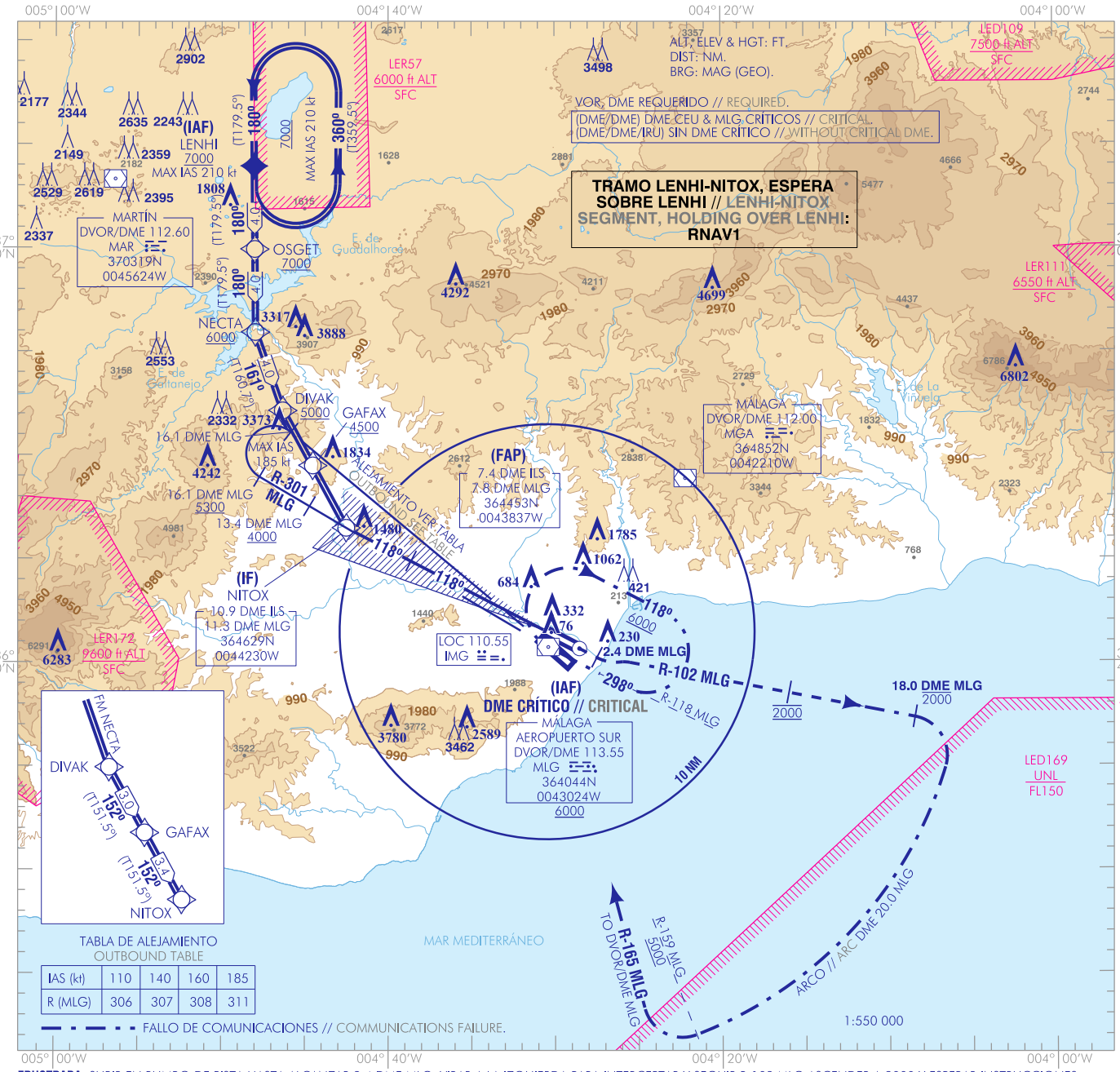



TABLA DE ALEJAMIENTO
OUTBOUND TABLE

| IAS (kt) | 110 | 140 | 160 | 185 |
|---|---|---|---|---|
| R (MLG) | 306 | 307 | 308 | 311 |

**FRUSTRADA**: SUBIR EN RUMBO DE PISTA HASTA ALCANZAR 2.4 DME MLG. VIRAR A LA IZQUIERDA PARA INTERCEPTAR Y SEGUIR R-102 MLG ASCENDER A 2000 Y ESPERAR INSTRUCCIONES ATC.

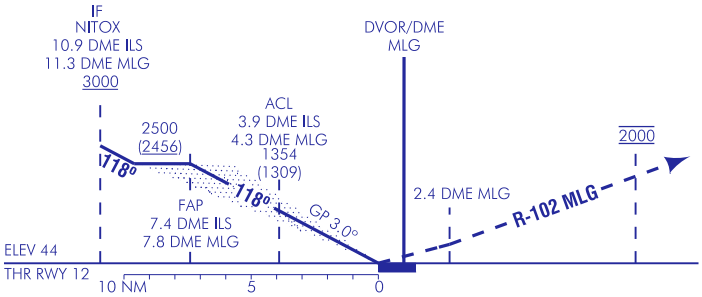
**FRUSTRADA FALLO DE COMUNICACIONES**: SUBIR EN RUMBO DE PISTA HASTA ALCANZAR 2.4 DME MLG. VIRAR A LA IZQUIERDA PARA INTERCEPTAR Y SEGUIR R-102 MLG, ASCENDER HASTA ALCANZAR 18.0 DME MLG A 2000 O INFERIOR. VIRAR A LA DERECHA PARA SEGUIR ARCO 20.0 DME MLG HASTA CRUZAR R-159 MLG A 5000 O SUPERIOR. VIRAR A LA DERECHA PARA INTERCEPTAR Y SEGUIR R-165 MLG DIRECTO A DVOR/DME MLG PARA INTEGRARSE A LA ESPERA A 6000.

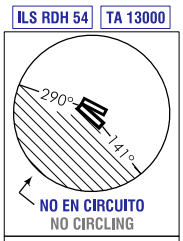
**MISSED APCH**: CLIMB ON RUNWAY HEADING TO REACH 2.4 DME MLG. TURN LEFT TO INTERCEPT AND FOLLOW R-102 MLG CLIMB AT 2000 AND AWAIT ATC INSTRUCTIONS.

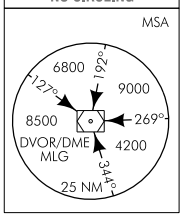
**MISSED APCH COMMUNICATIONS FAILURE**: CLIMB ON RUNWAY HEADING TO REACH 2.4 DME MLG. TURN LEFT TO INTERCEPT AND FOLLOW R-102 MLG, CLIMB TO REACH 18.0 DME MLG TO 2000 OR BELOW. TURN RIGHT TO FOLLOW ARC 20.0 DME MLG TO CROSS R-159 MLG TO 5000 OR ABOVE. TURN RIGHT TO INTERCEPT AND FOLLOW R-165 MLG DRECT TO DVOR/DME MLG TO JOIN THE HOLDING TO 6000.

NOTAS:
- CONTROL DE VELOCIDAD:
1. SI NO SE RECIBEN INSTRUCCIONES DIFERENTES DEL ATC, CRUZAR IF A IAS 190 kt, 8.0 DME ILS A IAS 180 kt Y 4.0 DME ILS A IAS 150 kt.
2. SI NO PUEDE CUMPLIR, NOTIFÍQUELO AL ATC EN PRIMERA COMUNICACIÓN.

NOTES:
- SPEED CONTROL:
1. IF ANY DIFFERENT INSTRUCTIONS ARE NOT RECEIVED BY ATC, CROSS IF AT IAS 190 kt, 8.0 DME ILS TO IAS 180 kt AND 4.0 DME ILS TO IAS 150 kt.
2. IN THE EVEN TO BEING UNABLE TO COMPLY WITH IT, NOTIFY IT TO ATC AT THE FIRST COMMUNICATION.




NO OFZ RWY 12

**HGT REF ELEV THR RWY 12**

| OCA/H | | A | B | C | D |
|---|---|---|---|---|---|
| STA | CAT I | 239 (195) | 251 (207) | 259 (215) | 270 (226) |
| | | | | | |
| | | | | | |
| En circuito (H) sobre / Circling (H) over 52 | | 650 (600) | 1000 (950) | 1480 (1430) | 2230 (2180) |

| GS | kt | 80 | 100 | 120 | 140 | 160 | 180 |
|---|---|---|---|---|---|---|---|
| FAP-THR: 7.4 NM | min:s | 5:33 | 4:26 | 3:42 | 3:10 | 2:46 | 2:28 |
| FAP-MAPT: | min:s | | | | | | |
| ROD: 5.2 % | ft/min | 425 | 531 | 637 | 743 | 849 | 955 |

| ALT/HGT DME (ILS) FNA | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
| | | | | | | 2380 (2330) | 2050 (2000) | 1720 (1670) | 1390 (1350) | 1070 (1020) | 740 (700) | 420 (380) |

CAMBIOS: RENUMERACIÓN, OBSTÁCULO.
CHANGES: RENUMBERING, OBSTACLE.

## MÁLAGA/Costa del Sol AD

REQUISITOS DE LA BASE DE DATOS AERONÁUTICA
AERONAUTICAL DATABASE REQUIREMENTS

### PROCEDIMIENTOS DE APROXIMACIÓN POR INSTRUMENTOS // INSTRUMENT APPROACH PROCEDURES

ILS Y RWY 12

| PUNTO<br>POINT | LAT | LONG | AZIMUT VERDADERO<br>TRUE BEARING | DISTANCIA DME<br>DME DISTANCE<br>(NM) |
|---|---|---|---|---|
| DVOR/DME MLG (IAF) | 364043.5N | 0043023.9W | - | - |
| NITOX (IF) | 364629.0N | 0044230.3W | 297.58º (LOC IMG) | 10.90 DME ILS<br>11.30 DME MLG |
| FAP | 364452.9N | 0043837.2W | 297.58º (LOC IMG) | 7.39 DME ILS<br>7.80 DME MLG |
| Aproximación final de no precisión - Pendiente (Ángulo de descenso)<br>Precision final approach - Slope (Descent angle) | | | | 5.24% (3.00º) |

| COORDENADAS WAYPOINTS // WAYPOINTS COORDINATES | |
|---|---|
| WPT | COORD |
| LENHI (IAF) | 370356.1N 0044803.3W |
| OSGET | 365955.8N 0044800.6W |
| NECTA | 365555.5N 0044758.0W |
| DIVAK | 365208.7N 0044619.3W |
| GAFAX | 364930.2N 0044432.4W |
| NITOX (IF) | 364629.0N 0044230.3W |

| DME CRÍTICO // CRITICAL DME | | |
|---|---|---|
| INSTALACIÓN (ID)<br>FACILITY (ID) | FREQ / CH | COORD |
| CEUTA (CEU) | -<br>CH124Y | 355333.0N 0051819.0W |

NOTAS APLICABLES A TODAS LAS APROXIMACIONES:
- RNAV 1.
- CONTROL DE VELOCIDAD:
  a) Si no se reciben instrucciones diferentes del ATC:
     • cruzar IF a IAS 190 kt,
     • 8.0 DME ILS a IAS 180 kt, y
     • 4.0 DME ILS a IAS 150 kt.
  b) Si no puede cumplir, notifíquelo al ATC en primera comunicación.

NOTES APPLICABLE TO ALL APPROACH:
- RNAV 1.
- SPEED CONTROL:
  a) If any different instructions are not received by ATC:
     • cross IF at IAS 190 kt,
     • 8.0 DME ILS at IAS 180 kt and
     • 4.0 DME ILS at IAS 150 kt.
  b) In the even to being unable to comply with it, notify it to ATC at the first communication.

DESCRIPCIÓN TABULAR DEL PROCEDIMIENTO
PROCEDURE TABULAR DESCRIPTION

| Número de serie<br>Serial number | Descriptor de trayectoria<br>Path Terminator | Identificador de punto de recorrido<br>Waypoint identifier | Sobrevuelo<br>Fly-over | Curso / Derrota<br>Course/Track<br>ºM (ºT) | Variación magnética<br>Magnetic variation | Distancia<br>Distance<br>(NM) | Dirección de viraje<br>Turn direction | Altitud<br>Altitude<br>(ft) | Velocidad<br>Speed<br>(kt) | VPA/TCH<br>(º/ft) | Especificación de navegación<br>Navigation specification |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **LENHI (IAF)**<br>(DME/DME) DME CEU CRÍTICO // CRITICAL.<br>(DME/DME/IRU) SIN DME CRÍTICO // WITHOUT CRITICAL DME. | | | | | | | | | | | |
| 001 | IF | LENHI | - | - | +0.6 | - | - | +7000 | -210 | - | RNAV1 |
| 002 | TF | OSGET | - | 180 (179.5) | +0.6 | 4.0 | - | +7000 | - | - | RNAV1 |
| 003 | TF | NECTA | - | 180 (179.5) | +0.6 | 4.0 | - | +6000 | - | - | RNAV1 |
| 004 | TF | DIVAK | - | 161 (160.7) | +0.6 | 4.0 | - | +5000 | - | - | RNAV1 |
| 005 | TF | GAFAX | - | 152 (151.5) | +0.6 | 3.0 | - | +4500 | - | - | RNAV1 |
| 006 | TF | NITOX | - | 152 (151.5) | +0.6 | 3.4 | - | +3000 | - | - | RNAV1 |

| CIRCUITOS DE ESPERA // HOLDING PATTERN | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Descriptor de trayectoria Path Terminator | Identificador de punto de recorrido Waypoint identifier | Sobrevuelo Fly-over | Curso/Derrota Course/Track ºM (ºT) | Variación magnética Magnetic variation | Distancia/ Tiempo de alejamiento Distance/ Outbound time | Dirección de viraje Turn direction | Altitud MNM/ MNM altitude (ft) | Altitud MAX MAX altitude (ft) | Velocidad Speed (kt) | Especificación de navegación Navigation specification |
| HM | LENHI | - | 180 (179.5) | +0.6 | 1 MIN | L | +7000 | - | -210 | RNAV1 |

INTENCIONADAMENTE EN BLANCO
INTENTIONALLY BLANK